■ Kitchmatics

AI-Powered Autonomous Restaurant System

Multi-Robot Fleet Management | Computer Vision | Voice AI

■ Agenda

Part 1-3

- ■■■■ ■■, ■■■ ■■■■, FMS ■■

Part 4-5

- ■■■ ■■, ■■ ■■ ■■ (4■)

Part 6-11

- AI ■■, ■■■, GUI, ■■■, ■■, ■■

■ ■ ■■

■ ■■

- 6■ ■■, 8■ ■■

■■ ■■

- FMS/Backend: ■■ ■■ ■■■
- Navigation: ■■ ■■■■■, SLAM
- AI Integration: YOLO, ■■■■
- HW/Embedded: ■■■ ■■

[illegible]

[illegible]

- [illegible]
- [illegible]

[illegible]

- [illegible] → [illegible] → [illegible]
- [illegible]
- AI [illegible]

■ ■■■■ ■■

1. ■■ ■■ ■■

- 3■ ■■■ ■■ ■■ ■■

2. End-to-End ■■■

- ■■ ■■■■ ■■ ■■■■

3. AI ■■ ■■

- ■■■ ■■■■ ■■

4. ■■ ■■

- ■■■ ■■ ■■

[illegible]

[illegible]

- 10,225+ Lines of Code
- 155 Test Cases
- 23 ROS2 Nodes

[illegible]

- 3 Mobile Robots (TurtleBot3)
- 2 Robot Arms (MyCobot 280)
- 5 Domain IDs

Core Platform

- ROS2 Humble, CycloneDDS, Nav2, Python 3.10+

AI/ML

- YOLOv8, OpenAI Whisper, GPT-4

Backend & Frontend

- FastAPI, PostgreSQL, PyQt5, TCP/JSON

Part 2

System Architecture

■ ■■ ■■■ ■■

■■ ■■■■

- Customer GUI (Kiosk) - ■■ ■■
- Admin GUI - ■■■ ■■■■
- AI Server - YOLO/Voice
- FMS - ■■ ■■
- Mobile Robots - pinky1, pinky2, pinky3
- Robot Arms - ■■

■ ■■■■ ■■

Mobile Robots (3■)

- TurtleBot3 Burger, LiDAR, Raspberry Pi 4

Robot Arms (2■)

- MyCobot 280, 6■, 250g ■■■■

Master PC

- Ubuntu 22.04, ROS2 Humble

Master PC (192.168.1.3)

- FMS Node, AI Server

Mobile Robots

- pinky1: 192.168.1.7 (Domain 11)
- pinky2: 192.168.1.6 (Domain 12)
- pinky3: 192.168.1.11 (Domain 13)

Robot Arm: 192.168.1.4 (Domain 20)

■ ROS2 Multi-Domain ■■

Domain ID ■■

- Domain 25: FMS (Master)
- Domain 11: pinky1
- Domain 12: pinky2
- Domain 13: pinky3
- Domain 20: Robot Arms

Why Multi-Domain?

- ■■ ■■■ ■■ ■■

■ Domain Bridge ■■

■■ ■■■

- /cmd_vel (Domain 11) → /pinky1/cmd_vel (Domain 25)
- /odom (Domain 11) → /pinky1/odom (Domain 25)

■■

- domain_bridge.yaml■■ ■■ ■■

■ Clean Architecture ■■

■■■ ■■

- Presentation: FMS Node, GUI
- Application: OrderHandler, FleetController
- Domain: Order, Robot Entities
- Infrastructure: ROS2, TCP, Database

■■■ ■■

- ■■ → ■■■ ■■■ ■■

Part 3

FMS Deep Dive

Fleet Management System ■■ ■■

■ FMS ■■

■■ ■■

- ■■ ■■ ■ ■■
- ■■ ■■ ■ ■■ ■■
- ■■-■■ ■■■
- ■■ ■■ ■ ■■

■■ ■■

- fms_node.py, order_handler.py
- fleet_controller.py, error_detector.py

■ FMS Node ■■

■■ ■■■■

- Application Layer: OrderHandler, FleetController
- Infrastructure: GUITCPServer (port=9000)
- ROS2: Navigation Action Clients

Dependency Injection

- ■■■■ ■■■ ■■

■ Order Handler

■■

- ■■ Workflow ■■
- ■■ ■■ ■■

■■ ■■

- RECEIVED → COOKING → LOADING
- → LOADED → DELIVERING
- → ARRIVED → COMPLETED

■ Fleet Controller

■■ ■■

- IDLE, MOVING, BUSY, ERROR

■■ ■■■■

- ■■■■: IDLE ■■ ■■
- ■■ ■■: ■■■ ■■■ ■■

■ Order State Machine

■■ ■■

- RECEIVED: ■■ ■■
- COOKING: ■■ ■ + ■■ ■■
- LOADING: ■■ ■■ ■■
- LOADED: ■■ ■■
- DELIVERING: ■■■■ ■■
- ARRIVED: ■■, ■■ ■■ ■■
- COMPLETED: ■■

■■ Error Detector

■■ ■■

- Navigation Timeout: ■■■ ■■■
- Robot Stuck: ■■ ■ ■■
- Communication Loss: ■■ ■■

■■ ■■

- RETRY, SKIP, MANUAL

Part 4

Mobile Robots

■ Navigation Stack

Nav2 ■■■■

- Nav2 BT Navigator: Behavior Tree
- Planner (NavFn): ■■ ■■
- Controller (DWB): ■■ ■■
- Costmap 2D: ■■■ ■

■ AMCL ■■ ■■

■■

- Particle Filter ■■
- LiDAR + Odometry ■■

■■ ■■■■

- min/max_particles: 500/2000
- update_min_d: 0.2m

■ Waypoint ■■

■■■ Waypoints

- kitchen (point13): ■■ ■■
- table1 ~ table4: ■■ ■■
- pinky1_spot ~ pinky3_spot: ■■ ■■

■■

- waypoints.yaml■■ ■■ ■■

Part 5

Problems & Solutions

■■ Problem #1: DDS Discovery ■■

■■

- Master PC■■ ros2 topic list ■■
- ■■■ ■■■ ■■ ■■■ ■■
- SSH, ping■ ■■ ■■

■■■ ■

- ■■■■■ ■■■■ ■■
- ROS2■ ■■ ■■

Problem #1: ■■ ■■

■■■ ■■

- 1. ■■■■ ■■ ■■ → ■■ ■■
- 2. Domain ID ■■ → ■■ ID
- 3. Wireshark → Multicast ■■■!
- 4. Router ■■ → Multicast ■■ ■■

■■ Problem #1: ■■ ■■

ROS2 DDS ■■ ■■

- ■■ ■■■ Multicast UDP ■■
- ■■: 239.255.0.1, ■■: 7400-7500

■■■

- WiFi AP■ Multicast ■■/■■

✓ Solution #1: Unicast Discovery

■■ ■■

- CycloneDDS Unicast Peer Discovery

■■

- XML ■■■ Peer IP ■■
- <Peer address='192.168.1.7'/>

■■ ■■

- RMW_IMPLEMENTATION=rmw_cyclonedds_cpp

✓ Solution #1: ■■

CycloneDDS XML

- cyclonedds_main.xml■ ■■ Peer ■■
- ■ ■■■ Master PC■ Peer■ ■■

■■

- ■■ WiFi ■■■■ ■■■ ■■

✓ Solution #1: ■■

Before (Multicast)

- WiFi AP■■ ■■ → ■■ ■■ ■■

After (Unicast)

- ■■ IP-to-IP ■■ → ■■ ■■ ■■

■■

- ■■ WiFi ■■■■ ROS2 ■■ ■■

■■ Problem #2: Topic Name Collision

■■

- ■■ ■■■ /cmd_vel, /odom ■■
- ■■■ ■■, ■■ ■■

■■

- ■■■ ■■■ ■■ ■■
- Fleet ■■ ■■■

■■ Problem #2: ■■

■■■ ■■

- ■ ■■■ ■■■ ■■
- ■ ■■ ■■ ■■
- ■ ■■ ■■

■■■ ■■

- ■ TurtleBot3 ■■■ ■■ ■■■
- ■ Nav2 ■■■ ■■■■

✓ Solution #2: Multi-Domain Architecture

■■ ■■■■

- ■ ■■■ ■■ Domain■■ ■■
- Domain Bridge■ ■■

■■

- Domain 11 (pinky1): /cmd_vel
- Domain 25 (FMS): /pinky1/cmd_vel

✓ Solution #2: Domain Bridge ■■

YAML ■■

- from_domain: 11, to_domain: 25
- topic: /cmd_vel → /pinky1/cmd_vel

■■

- ros2 run domain_bridge domain_bridge

✓ Solution #2: ■■

FMS Domain ■■ ■■

- /pinky1/cmd_vel, /pinky1/odom
- /pinky2/cmd_vel, /pinky2/odom

■■

- ■ ■■■ ■■■ ■■ ■■
- ■ ■■ ■■

■■ Problem #3: ■■-■■ ■■■

Case A: ■■ ■■ ■■

- ■■ ■■ ■■ ■ $\rightarrow$ ■■

Case B: ■■ ■■ ■■

- ■■ ■■ ■■ ■ $\rightarrow$ ■■ ■■

■■ Problem #3: ■■■ ■■

Approach 1: ■■■

- 30■ ■■ ■ ■■ ■■
- ■ ■■ ■■■

Approach 2: ■■

- ■■■ ■■ ■■
- ■ CPU ■■, ■■ ■■

✓ Solution #3: Event-Driven State Machine

■■ ■■

- ■■■ ■■■■ ■■
- AND ■■■■ ■■ ■■

■■

- cooking_complete + robot_at_kitchen
- ■ ■ True■ LOADED■ ■■

Solution #3: ■■ ■■■■■

COOKING ■■■■ ■■

- Event A: Robot Arrived
- Event B: Cooking Done

AND ■■

- A AND B ■■ ■ LOADED ■■

✓ Solution #3: ■■

■■ ■■

- Race Conditions: 0
- ■■■ ■■■: 100%
- ■■ ■■ ■■: -40%

■■ ■■

- ■■■ ■■, ■■■ ■■

■■ Problem #4: GUI-FMS ■■

■■■■

- ■■ ■■, ■■ ■■ (Push)
- ■■ ■■, ■■■ ■■■■

■■■■

- ■ GUI■ ROS2 ■■ ■■
- ■ TCP/IP ■■ ■■

✓ Solution #4: TCP/JSON Protocol

■■■ ■■

- 4-byte length header + JSON

■■■ ■■

- new_order: ■■
- delivery_notification: Push
- delivery_complete: ■■ ■■

Solution #4: ■■ ■■

GUITCPServer

- Thread-per-client ■■
- Handler ■■■■ ■■
- broadcast()■ Push ■■

✓ Solution #4: ■■

■■

- ■■ ■■: <100ms
- Push ■■: <50ms

■■

- ■ ROS2 ■■ ■■■ ■■
- ■ ■■■ ■■■■■ ■■

Part 6

AI Integration

AI ■■ ■■

■ AI ■■ ■■

YOLO Server

- YOLOv8, ■■ ■■ ■■, FastAPI

Voice Server

- Whisper STT, GPT-4 Function Calling
- OpenAI TTS

■ YOLOv8: ■■ ■■■

■■■■

- ■, ■■, ■■, ■■■, ■■■
- ■ 350~500■

■■ ■■

- YOLOv8n, 100 epochs, 640x640

■ YOLOv8: ■■

API

- POST /detect
- ■■: ■■■, ■■: detections

FMS ■■

- ■■■ ■■ ■■ ■■ ■ ■■ ■■

□ □□ □□ □□□□□

□□ □□

- Microphone → Whisper → Text
- → GPT-4 Function Calling → JSON
- → TTS → Speaker

□□

- '□□□□ □ □' → {menu, qty}

■ GPT-4 Function Calling

Function ■■

- name: create_order
- parameters: menu_name, quantity

■■

- ■■■ → ■■■■ JSON

Part 7

Robot Arm

■ ■■■ ■■■■

MyCobot 280

- 6■, 250g ■■■■, 280mm ■■

End Effector

- Gripper, 0-70mm

■ FMS ■■

■■ ■■■■■

- /arm/command: ■■ ■■
- /arm/status: ■■ ■■

■■ ■■

- job_id, operation, order

RecipeExecutor

- YAML ■■■ ■■
- step■ ■■ ■■

■■■ ■■

- ■■■ → ■■ → ■■ → ■■■ → ■■■

Part 8

GUI Development

■ Customer GUI

■■ ■■

- ■■ ■■, ■■■■, ■■
- ■■, ■■ ■■

■■

- PyQt5, TCP/JSON

□ □□ □□□

□□

- 1. □□ □□
- 2. □□□□
- 3. □□
- 4. □□ □□
- 5. □□ □□
- 6. □□ □□

■ Admin GUI

■■■■

- Fleet ■■, ■■ ■■, ■■ ■■

■■

- ■■ ■■, ■■ ■■, ■■ ■■

Part 9

Testing Strategy

- Integration Tests: E2E Flow
- Component Tests: FMS, Nav, AI
- Unit Tests: Functions & Classes

■ Skip Mode

■■

- ■■■ ■■ ■■ ■■■ ■■

■■

- skip_mode ■■■■
- ■■■■ ■■ ■■■■■

[illegible]

[illegible]

- 155 Test Cases
- 78% Code Coverage
- 100% Critical Path

[illegible]

- Order Handler: 92%
- State Machine: 100%

Part 10

Performance & Scalability

■ ■■ ■■

Latency

- ■■: <100ms, Push: <50ms
- Navigation: <200ms, YOLO: <100ms

Throughput

- ■■ ■■: 10+
- ■■ ■■: ~60s

Core Tables

- orders, order_items, menu_items
- robots, delivery_history

■■

- PostgreSQL, ■■■, ■■■ ■■■

■ ■■■ ■■

■■ ■■

- ■■ ■■: Domain ID■ ■■
- FMS ■■■ ■■

■■ ■■

- Redis ■■, AsyncIO

Part 11

Conclusion

■ ■■■ ■■

ROS2/DDS

- WiFi → Unicast ■■
- Multi-Domain■■ ■■

Architecture

- Clean Architecture
- State Machine

□ □□□□ □□

□□ □□ □□□

- □□□□□ □□ □□

□□□

- API □□, □□□□ □□□□□

□□□□□

- Skip Mode□ HW □□ □□

■ ■■ ■■: ■■

3■■ ■

- ■■ 5■ ■■
- ■ Admin Dashboard

6■■ ■

- Mobile App
- ■■ ■■■■

■ ■■ ■■: ■■

1■ ■

- ■■ ■■ ■■■■■
- ■■ ■■ ■■■

2■ ■

- ■■ ■■ ■■
- ■■ ■■ ■■

■ ■■

■■ ■■

- ■ ■■ ■■ ■■ (3■)
- ■ End-to-End ■■■
- ■ AI ■■ ■■
- ■ ■■ ■■

■■ ■■

- DDS Discovery ■■
- Multi-Domain Architecture

■■■■■

Q&A

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